Sulu State University
COLLEGE OF COMPUTING STUDIES
Capitol Site, Bangkal, Patikul, Sulu

AN INTELLIGENT MACHINE LEARNING SYSTEM FOR PROGRAMMING DIFFICULTY ASSESSMENT AND CODING ENGAGEMENT ENHANCEMENT

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DEDICATION

Innalhamdulillah hi rabbil Alamin thanks be to Allah, this study is wholeheartedly dedicated to our beloved families, whose unconditional love, patience, sacrifices, encouragement, and continuous support have served as our greatest source of strength throughout our academic journey. Their understanding and motivation inspired us to persevere despite the challenges, difficulties, and pressures encountered during the development of this study. We also dedicate this work to our instructors, advisers, and mentors who generously shared their knowledge, guidance, experiences, and constructive suggestions, particularly in the areas of programming, information technology, research, and system development. Their dedication to teaching has encouraged us to continuously improve our skills, think critically, solve problems, and explore the meaningful applications of technology. This study is especially dedicated to the students and academic community of **Sulu State University**, particularly those pursuing Information Technology, Computer Science, Computer Engineering, and other computer-related programs. We hope that this study, entitled **"An Intelligent Machine Learning System for Programming Difficulty Assessment and Coding Engagement Enhancement,"** can contribute to the continuous improvement of programming education by providing an intelligent tool that can help learners identify coding errors, understand programming difficulties, recognize the programming language used, receive appropriate recommendations, and monitor their assessment results. We dedicate this work to

students who continue to learn programming despite encountering difficulties in understanding syntax, logic, algorithms, and problem-solving concepts, and who strive to develop their skills through continuous practice and determination. May this study encourage students to view programming not merely as a requirement for graduation, but as an important skill that can develop creativity, logical thinking, innovation, and the ability to create solutions to real-world problems. Finally, this study is dedicated to everyone who believes in the potential of students to learn, create, innovate, and use technology for meaningful purposes. May this work serve as a small contribution to the academic community of **Sulu State University** and inspire future students and researchers to continue exploring intelligent technologies that can support learning and improve the educational experience.

TABLE OF CONTENTS

TITLE PAGE I
APPROVAL SHEET II
DEDICATION III
TABLE OF CONTENT V
LIST OF FIGURES VII
LIST OF TABLES VIII
EXECUTIVE SUMMARY IX
KEYWORDS X

CHAPTER I: INTRODUCTION 1
Project Context 2
Purpose and Description 5
Objectives 6
General Objective 6
Specific Objectives 7
Significance of the Project 7
Scope and Delimitation 9

CHAPTER II: REVIEW OF RELATED LITERATURE / SYSTEM TECHNICAL BACKGROUND 11
Related Literature 12
System Technical Background 12
Foreign Studies 13
Local Studies 14

CHAPTER III: METHODOLOGY 16
Model / SDLC 17
Data Gathering 18
Requirements Analysis 20
Functional Requirements 20
Non-Functional Requirements 21
Requirements Documentation 21
Hardware Requirements 22
Software Requirements 22

Network Requirements 23
Development & Deployment Requirements 24
Performance & capacity planning 25
Optional / future-proofing 25
Foundation of Software Development 26
Detailed Design 27
Design of Software, Systems, Product, and/or Processes 22
System Architectural Foundation 22
System Layered Architecture 23
Detailed Design 27
Presentation Layered Architecture 28
Conceptual Framework 29
System Process Flow 30
Algorithms 31
Machine Learning 31
Capstone Element 35
Testing & Validation 37

CHAPTER IV: RESULTS AND DISCUSSION 38
System Development Results 38
Student Registration and Profile Module 40
Coding Activity Module 40
Report Module 41
System Outputs 42
Performance Testing 43
User Acceptance Testing 44
Evaluation of the System 44
Functionality 44
Usability 44
Reliability 45
Efficiency 45
Security 45
Discussion of Finding 45

CONCLUSION AND RECOMMENDATION 46
Conclusion 46
Recommendations 47
REFERENCES 50
APPENDICES 54
Appendix A. Relevant Source Code 54
Appendix B. User's Guide 64
Appendix C. Screen Layouts 79
Appendix D. Test Results 85
Appendix E. Pictures Showcasing Data Gathering 92
Appendix F. Curriculum Vitae 93

LIST OF FIGURES

Figure 0. Sustainable Development Goals 8
Figure 1. Agile Development Life Cycle 17
Figure 2. The Foundation of System Architecture 26
Figure 3. Detailed Design 27
Figure 4. The Layered Architecture 28
Figure 5. The Foundation of System Architecture 26
Figure 6. The Conceptual Frame Work (IPO) 29
Figure 7. The System Flow Diagram 30
Figure 8. The Student Registration Module 40
Figure 9. The Coding Activity Module 41
Figure 10. The Report Module 42
Figure 11. The System Output Module 43
Figure 12. The Overall System Work Flow 77
Table 13. Screen Layout User Start 79
Table 14. Screen Layout Admin Start 81

LIST OF TABLES

Table 1. Functional Requirements 20
Table 2. Non – Functional Requirements 21
Table 3. Troubleshooting 75
Table 4. Student Management Result 85
Table 5. Coding Assessment Test Results 86
Table 6. Report Test Results 87
Table 7. Overall Functional Testing 88
Table 8. Coding Analysis Results 88
Table 9. Recommendation Testing 89
Table 10. Usability Test Results 90

EXECUTIVE SUMMARY

The study entitled **"An Intelligent Machine Learning System for Programming Difficulty Assessment and Coding Engagement Enhancement"** focuses on the development of a machine-learning-based system intended to assist students and instructors in assessing programming difficulties and improving coding engagement. Programming is an essential skill in Information Technology and computer-related programs; however, students, particularly beginners, may experience difficulties in understanding programming concepts, identifying errors, and developing effective problem-solving strategies.

The proposed system consists of three major features: **Student Management, Coding Assessment, and Report**. The Student Management feature allows authorized users to organize and maintain student information. The Coding Assessment feature serves as the primary intelligent component of the system. It analyzes submitted source code, distinguishes the programming language being used, evaluates whether the code is correct or contains errors, and provides recommendations based on the detected coding problem. The system is intended to transform coding assessment from a simple correct-or-incorrect evaluation into a more informative process that can help students understand and improve their programming performance. Automated programming assessment and intelligent feedback have been explored in computer science education as approaches for providing scalable assessment and supporting programming practice (Ala-Mutka, 2005; Combéfis, 2022; Lan, 2026).

The Report feature provides organized assessment information that can be used to review students' coding performance, identified difficulties, and assessment results. By combining automated code analysis, recommendations, and organized reporting, the proposed system aims to provide students with timely feedback while giving instructors useful information for monitoring programming performance. The system is therefore designed as an educational support tool that can contribute to more responsive programming instruction and greater student engagement in coding activities.

KEYWORDS

Machine Learning; Programming Difficulty Assessment; Coding Assessment; Automated Code Analysis; Programming Language Detection; Error Detection; Code Correctness; Intelligent Recommendation; Coding Engagement; Student Management; Programming Education; Learning Analytics